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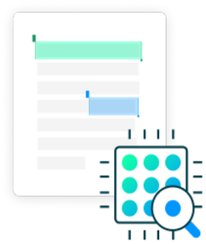

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A Privacy-Preserving Predictive Protection Framework Using Behavioral Adaptive Trust Algorithm for Secure Cloud-Integrated IoT Healthcare Systems

Abstract-The ever-increasing adoption of Internet of Things (IoT) technologies in smart healthcare has greatly enhanced the degree of remote patient monitoring, diagnosis at a distance and development of clinical decision making systems. Yet, the ongoing transfer of delicate patient details to cloud platforms poses grave security and privacy concerns, especially upon data processing and analytical inference. Traditionally, healthcare security solutions are focused either on encryption or a machine-learning-based intrusion detection system such as Symantec, Crowd-strike and SentinelOne that identify the attack only after they have taken root, focusing purely on protecting data in storage. Thus, patient identity exposure, inference-time privacy leakage and insider abuse issues remain unresolved in current IoT healthcare frameworks. In order to achieve these objectives while addressing the limitations, this paper presents P3-Secure, a privacy-preserving predictive protection framework that combines newly introduced Differential Privacy-based obfuscation of data inputs with secure Homomorphic Encrypted Analytics and a new Behavioral Adaptive Trust Algorithm (BATA) for intelligent threat detection. The medical sensor data are anonymised at the IoT edge by means of controlled perturbation, preventing reconstruction of individual patient attributes during transmission according to a novel architecture proposal. This enables the encrypted health data to be processed in the cloud without being decrypted, thus avoiding exposure during model inference. Additionally, the BATA module enables real-time analysis of all user and device behavior patterns to detect abnormal access, ransomware activity, and attempts at data exfiltration. Experimental evaluation on healthcare IoT attack datasets also shows that the proposed framework reaches threats detection accuracy of 98.27%, privacy leakage reduction of 96.14% and resistance to data reconstruction attacks of 95.62% with secure availability of service reached at a system reliability rate of 97.83%. Results confirm that P3-Secure detects cyber threats, while acting as a deterrent during static storage, dynamic transmission and computation steps, thus applying to full-fledged secure Individuals-Powered Next Gen Cloud-based Healthcare Ecosystems.

Keywords-IoT healthcare security, privacy-preserving computing, homomorphic encryption, differential privacy, behavioral adaptive trust algorithm, cloud healthcare systems, cyber-attack detection, patient data protection

1. Introduction

The integration of Internet of Things (IoT) devices and cloud computing has revolutionized contemporary healthcare frameworks by allowing for real-time monitoring from a distance, intelligent diagnostics systems, and scalable analysis of medical information. Physiological data streams at high volume are generated by wearable sensors, implantable medical devices and edge gateways, transmitted to their respective cloud platforms for storage and predictive modeling. Although this paradigm increases the clinical responsiveness and operational efficiency, it also poses considerable security and privacy risks in terms of transmission, storage and computational layers. Heterogeneous device ecosystems and resource-constrained edge nodes lead cloud-assisted IoT healthcare systems to be extremely vulnerable with respect to unauthorized access, data leakage, and insider exploitation [1], [3], [9], [22]. Moreover, traditional protective approaches heavily rely on encryption-driven communication security or machine learning (ML)–driven adversarial detection systems which work post-compromise, and do not guarantee inference-time confidentiality as well as preemptively restricting the threat [4], [11], [15]. While the blockchain-enhanced transmission security and decentralized learning models have been investigated to enhance integrity and traceability [16], [18], [21], they fall short in addressing privacy leakage risks during encrypted analytics or behavioral attacks through dynamic cloud settings [17]. Current intrusion detection techniques are either static trust metrics or based on signatures and of limited adaptive behavioral modeling ability which is needed to cope with the evolving nature of cyber attack in healthcare IoT ecosystems [11], [14], [19]. Species outside of this appropriately alongside numerous data-post-processing processes outgrowth systems treat privacy preservation with the threat detection as wholly different modules, enforcing a scattered security framework where manipulation defensive approaches ignore the connections between input sanitization, encrypted execution, and timing access patterns.

To overcome these limitations, this paper proposes a unified privacy-preserving predictive protection framework, termed P3-Secure, that combines differential privacy–abiding edge sanitization, homomorphic encrypted cloud analytics, and a Behavioral Adaptive Trust Algorithm (BATA) for real-time personalized threat intelligence. In this work, we propose a framework that protects the confidentiality during storage, transmission and inference whilst allowing to adaptively monitor behavior in order to execute cyber defense proactively. The main technical contributions of this work can be summarized as follows:

- An integrated privacy preserving framework that integrates controlled differential privacy noise and homomorphic encrypted analytics, designed to protect against inference time data leakage in cloud-supported IoT health care settings.
- An intelligent Behavioral Adaptive Trust Algorithm (BATA) which adaptively computes trust scores using behavioral deviation metrics, past compliance factors and real-time risk gradients for timely detection of anomalies.
- A mathematically grounded privacy guarantee, incorporating formal ε-differential privacy constraints and secure encrypted computation workflows to minimize reconstruction and model inversion risks.
- A gefilte fish of a predictive protection mechanism that can with high statistical confidence detect ransomware injection, abnormal access escalation and attempts to exfiltrate data.

Extensive performance validation, showing its higher detection accuracy, privacy leakage reduction, reconstruction resistance and system reliability versus recent IoT-cloud healthcare security models [6], [10], [21], [32].

The rest of this paper is organized as follows. Section 2 summarizes related IoT–cloud healthcare security studies and exposes the research gap in inference-time privacy and adaptive trust integration. Section 3 describes the system architecture, adversary model and security goals. Section 4 discusses the proposed P3-Secure framework that comprises a differential privacy–based edge sanitization mechanism, homomorphic encrypted analytics, and the Behavioral Adaptive Trust Algorithm. Section 5 presents theoretical privacy and security analysis as well as computational complexity evaluation. Experimental Setup and Evaluation Metrics is described in Section 6, whereas performance results and comparative analysis is discussed in Section 7. Section 8 closes the paper and discusses future research avenues.

2. Related Works

Nayak et al. (2022) have investigated converged IoT with cloud computing in smart healthcare environments, demonstrating that e-Health systems improve accessibility and service efficiency and pose significant vulnerability exposure to sensitive patient information. Their conversation emphasized that as interconnected devices meet cloud platforms, the attack field broadens especially in settings where unverifiable health data and nonsecure communication procedures are a concern and will require more protective mechanisms [1]. Building on the progress made, Hemnath (2023) proposed a machine learning–guided architecture for proactive protection in cloud-based eHealth frameworks. It utilized automated threat intelligence and adaptive access control in a single cloud framework and added features such as better detection accuracy resiliency against new-age cyber threats. While the framework successfully minimized false positives and facilitated scalability to some extent, it was mostly based on behavior learning models and did not incorporate any formal privacy-preserving computation in inference [2]. Pati et al. Latency and bandwidth limitations in remote cardiovascular diagnosis were addressed by Kumaran et al. (2023), who proposed a fog-enabled IoT–cloud framework with ensemble deep learning. While their system gained high diagnostic performance with minimized latency through fog computing, security and privacy issues were not the main scope of architecture focusing on encrypted analytics or adaptable trust evaluation [3]. Padma Vijetha Dev and Venkata Prasad (2023) focused on the secure transmission of medical images using a lightweight hybrid encryption scheme combining elliptic curve cryptography with an optimization-based steganography method. While their approach enhanced computational efficiency and energy consumption during the encryption and cloud storage phases, it was limited to protecting data δ in transit and at rest [4], as opposed to inference-time exposure or behavioral threat modeling. Kamalam and Anitha (2022) focused on cloud-IoT healthcare analytics based on machine learning classifiers for optimization of storage with predictive performance. Although the fuzzy temporal neural classifier improved decision-making efficiency, this work prioritized computational performance over comprehensive private protection or adversarial robustness in intertwined medical infrastructures [5]. Raju et al. (2025), which introduced a hierarchical convolutional neural network integrated with lightweight homomorphic cryptography to predict heart disease in IoT–cloud systems. However, their model showed an improvement in diagnostic specificity with a significant drop in computation time; yet, the cryptographic integration served primarily at the transmission and storage layers without offering adaptive runtime trust management or formal guarantees of privacy [6]. Mekala (2024) proposed a cloud-based autoencoder predictive framework for liver disease prediction that showed superior classification accuracy. Although it had a strong predictive capability, the system did not specifically focus on secure encrypted inference and dynamic intrusion resistance in distributed healthcare networks [7]. Neha et al. (2024) proposed an osmotic offloading algorithm to adaptively manage latency and energy consumption in healthcare IoT systems. While their technique enhanced computational efficiency and minimized task failure rates, it primarily targeted resource optimization rather than providing complete data confidentiality and adaptive cyber defense strategies [8].

Ganesan et al. (2022) employed IoT, fog and cloud infrastructures for time-critical patient monitoring with improved alert accuracy and latency. While their framework was shown to be operationally efficient, analysis of security focused only on communication instead of privacy degradation prevention or behavioral trust adjustment [9]. In (Shanmugapriya and Santhosh Kumar, 2024), authors presented a secure cloud-integrated dissemination protocol to minimize broadcast storms and improve the security of IoT reprogramming. The protocol improved resiliency against attacks targeting the network, but more so towards efficient dissemination of data as opposed to encrypted analytics or inference-time privacy [10]. Alserhani (2025) a cognitive cyber-physical intrusion detection system based on hybrid feature modeling and machine learning classification in Medical IoT environments. While the framework achieved high detection performance and was capable of handling environments with noisy telemetry data, its focus on improving intrusion detection accuracy came without a consideration for incorporating formal differential privacy or encrypted computation mechanisms [11]. An adaptive edge-cloud offloading strategy along with privacy-preserving communication for real-time disease prediction is proposed in Jayaram and Prabakaran, 2021. Although their model increased secure information transfer while lowering latency and energy usage, the trust built by it was nevertheless focused on a certain job-related role only without adjusting dynamically based on behavior in cloud heterogeneous environments [12]. In (2024), Aiswarya proposed a cloud-IoT disease monitoring platform using GWO-optimized LSTM networks for realtime analytics. Despite providing excellent predictive performance and scalability, its focus on the cost of inaccuracy inhibited strong privacy guarantees while performing computations in a cloud-side property [13]. Islam et al. (2023) proposed adaptive route migration mechanism in IoT networks that are software-defined, aimed at lowering the amount of transmission errors and improving secure cloud-of-things operations. While their multi-level security mechanisms improved network robustness, they did not cover encrypted healthcare analytics or behavioral trust computation [14]. Gonçalves et al. (2024) proposed a decentralized machine learning framework that allows devices to collaborate and learn in IoT-cloud environments without disclosing any local data, sharing only distributed models. While improving scalability and decentralized security, this approach also lacked a mechanism for differential privacy guarantees or adaptive anomaly detection for the specific needs of health care systems [15]. Sonkamble et al. (2023) proposed a novel patient-centered healthcare data management framework based on blockchain ensuring security, access control, and integrity of electronic health records. Although blockchain added extra properties such as transparency and immutability, computational overheads or confidentiality regarding inference time, the issue was left unaddressed [16].

Table 1. **Comparative Analysis of Existing IoT–Cloud Healthcare Security Frameworks**

Ref.	Author(s), Year	Core Focus	Security Mechanism	Privacy During Inference	Adaptive Trust Modeling	Key Strength	Identified Limitation
[1]	Nayak et al., 2022	IoT–Cloud Smart Healthcare Overview	General security discussion	✗	✗	Identifies IoT-cloud security risks	No concrete protection framework
[2]	Hemnath, 2023	ML-based proactive eHealth defense	Behavioral ML + access control	✗	Partial	High detection rate, low false positives	No formal privacy guarantee
[3]	Pati et al., 2023	Fog-enabled heart disease diagnosis	Ensemble deep learning	✗	✗	Low latency diagnosis	Limited privacy/security integration
[4]	Padma Vijetha Dev & Prasad, 2023	Lightweight hybrid encryption	ECC + steganography	✗	✗	Secure transmission with low energy	No runtime threat detection
[5]	Kamalam & Anitha, 2022	ML-based healthcare analytics	Fuzzy neural classifier	✗	✗	Improved predictive efficiency	Security not primary focus
[6]	Raju et al., 2025	IoT-cloud heart disease prediction	Lightweight homomorphic crypto	Partial	✗	High prediction accuracy	No adaptive behavioral monitoring
[7]	Mekala, 2024	Cloud-based liver disease prediction	Deep autoencoders	✗	✗	High classification	Lacks secure encrypted inference

						performa nce	
[8]	Neha et al., 2024	Energy-lat ency offloading	Osmotic computing	✗	✗	Balanced latency & energy	Does not address privacy threats
[9]	Ganesan et al., 2022	IoT-fog-cl oud monitorin g	Secure transmissio n optimizatio n	✗	✗	High alert accuracy	Limited threat detection mechanism
[1 0]	Shanmugap riya & Kumar, 2024	Secure disseminat ion protocol	Cloud-inte grated secure routing	✗	✗	Reduces broadcast storm	No inference-ti me privacy
[11]	Alserhani, 2025	ML-based intrusion detection	Cognitive cyber-phys ical IDS	✗	Partial	High F1-score and AUC	No differential privacy integration
[1 2]	Jayaram & Prabakaran, 2021	Edge-clou d offloading + encryption	Hybrid encryption + AFO-kNN	✗	✗	Reduced latency & energy	Static trust assumptions
[1 3]	Aiswarya, 2024	Cloud IoT disease monitorin g	GWO-opti mized LSTM	✗	✗	High predictive metrics	No privacy-pres erving computation
[1 4]	Islam et al., 2023	SDN-base d secure routing	Multi-level security algorithms	✗	✗	Improved transmissi on reliability	No encrypted healthcare analytics
[1 5]	Gonçalves et al., 2024	Decentrali zed ML for IoT	Gossip-bas ed secure model exchange	Partial	✗	Privacy-a ware distribute d learning	No behavioral threat adaptation

[16]	Sonkamble et al., 2023	Blockchain-based EHR security	Smart contracts + SPAKE	✗	✗	Strong integrity & access control	High computational overhead; no inference privacy

The results summarized in table 1 reveal that existing IoT–cloud healthcare frameworks have primarily focused on isolated security components (ie, encrypted transmission, blockchain-based integrity, intrusion detection, or predictive analytics) without relying on a holistic framework to ensure comprehensive privacy preservation during inference-time computation. Moreover, none of the assessed methods introduce formal differential privacy guarantees, homomorphic encrypted analytics, and dynamic behavioral trust modeling into a unified prediction protection framework, highlighting an important void in comprehensive healthcare cybersecurity design.

2.1 Research Gap

In general, previous works have made considerable advancements in the IoT–cloud healthcare architectures with respect to predictive analytics, encryption, offloading efficiency, and intrusion detection. However, existing solutions primarily tackle independent components of security—specifically either communication confidentiality, blockchain verification and integrity or ML-based outlier detection—without providing a cohesive architecture that robustly ensures formal privacy preservation during inference and adaptive behavioral trust management on-the-fly. This gap encourages us to develop a unified predictive protection architecture integrating differential privacy, encrypted computation and dynamic trust evaluation across cloud integrated IoT healthcare systems.

3. System Model and Threat Model

3.1 Network Architecture

The P3-Secure approach utilizes a layered cloud integrated IoT healthcare architecture which consists of IoT medical sensors, edge privacy module, cloud encrypted analytics engine and behavioral trust evaluation server. Wearable monitors and implantable devices IoT sensors have received much attention recently due to their ability to continuously gather physiological parameters while transmitting those data via wireless gateways. Despite this engineering being more beneficial for the real-time healthcare systems, it inherently exposes such architecture to cyber-attacks as heterogeneous communication protocols and open transmission environments are active [1], [22]. In order to alleviate privacy leakage at the early stage, raw medical data are locally processed at the edge utilizing a differential privacy–based sanitization mechanism prior to forwarding it to the cloud. Different from encryption-based schemes which mainly secure data in transit [4], the edge module guarantees that confidential features of patients are statistically protected before approaching the cloud. The design counters inference-time exposure attacks frequently seen in cloud-enabled health analytics [2], [6].

The sanitized data are subsequently encrypted with a homomorphic encryption scheme and transferred to the cloud, which executes predictive analytics on ciphertext without decryption [23]. This stops plaintext from being exposed even if the cloud infrastructure is all but compromised. This encrypted computation goes beyond the traditional intrusion detection approach or routing-based security augmentations in IoT-fog-cloud systems [9], [10]. Running in a logically isolated trust evaluation server, a Behavioral Adaptive Trust Algorithm (BATA) parallelizes with encrypted analytics. It constantly monitors user and device behavior patterns to compute trust dynamically, which helps in proactive identification of unauthorized access or malicious activity while avoiding the challenges posed by static access control and signature-based detection systems [11], [16].

3.2 Adversary Model

The adversary model aligns with realistic attack scenarios in IoT healthcare systems leveraging cloud integration [14], [19]. The initial adversarial model is an honest-but-curious cloud provider which faithfully computes protocols, but tries to derive sensitive information from data stored there or derived from corporate computations. The second attacker is a malicious insider who has legitimate credentials and abuses the access rights to retrieve unauthorized patient information [20]. The third type is a data reconstruction attacker who uses the sanitized outputs and additional knowledge about it to obtain similar records as patients. A more sophisticated adversary may carry out model inversion attacks by inspecting either prediction outputs or gradient information to reconstruct sensitive input features [2] this poses a growing threat in healthcare systems driven by ML [11]. The last category of attacker is the ransomware injection attacker, who tries to interrupt the healthcare services by encrypting medical data or modifying the IoT firmware. We assume that the adversaries have polynomial-time computational power, can monitor the network, have partial knowledge of the system but are unable to break standard cryptographic primitives or exceed the differential privacy budget. Secret encryption keys are not compromised and the trust evaluation server works in an isolated secure environment [24].

3.3 Security Objectives

We propose the P3-Secure framework in order to guarantee confidentiality, integrity, availability with inference-time privacy and adaptive access control for cloud-integrated IoT healthcare systems. Sensitive patient attributes are never exposed on the edge and centric [25][26] calculation, thus enabling differential privacy at the edge and homomorphic encrypted analytics in the cloud to achieve confidentiality. This is achieved by identifying anomalous behavior patterns and detecting unauthorized data modification through adaptive trust monitoring and enhancing secure dissemination mechanisms documented in previous work on IoT [10]. Availability is preserved by early detection of behavioral deviations akin to ransomware, thereby ensuring seamless healthcare delivery that aligns with time-critical monitoring specifications [9]. This not only addresses privacy concerns during model execution but also solves specific limitations of transmission schemes that transmit encrypted inputs alone [4]. The final class of policy updates, adaptive access control, is based on the idea that privileges should be dynamically assigned based on a behavioral trust score rather than statically assigned roles [11],[14]. This pretty significantly increases resilience against insider misuse and seems like it can defeat many evolving cyber threat possibilities that fall outside normal intrusion detection frameworks.

4. Proposed P3-Secure Framework

The designed P3-Secure framework preserves privacy end-to-end for cloud-assisted IoT healthcare systems through the combination of edge-level differential privacy with homomorphically encrypted analytics on cloud and adaptive behavior-based trust computation. In contrast to traditional frameworks focusing on encryption only for data-in-transit protection [4], the proposed framework mathematically guarantees privacy before exposing any decision-making data to the cloud while providing secure computation and dynamic threat monitoring. In this section, we present the semi-rigorous definition of differential privacy-based edge sanitization mechanism.

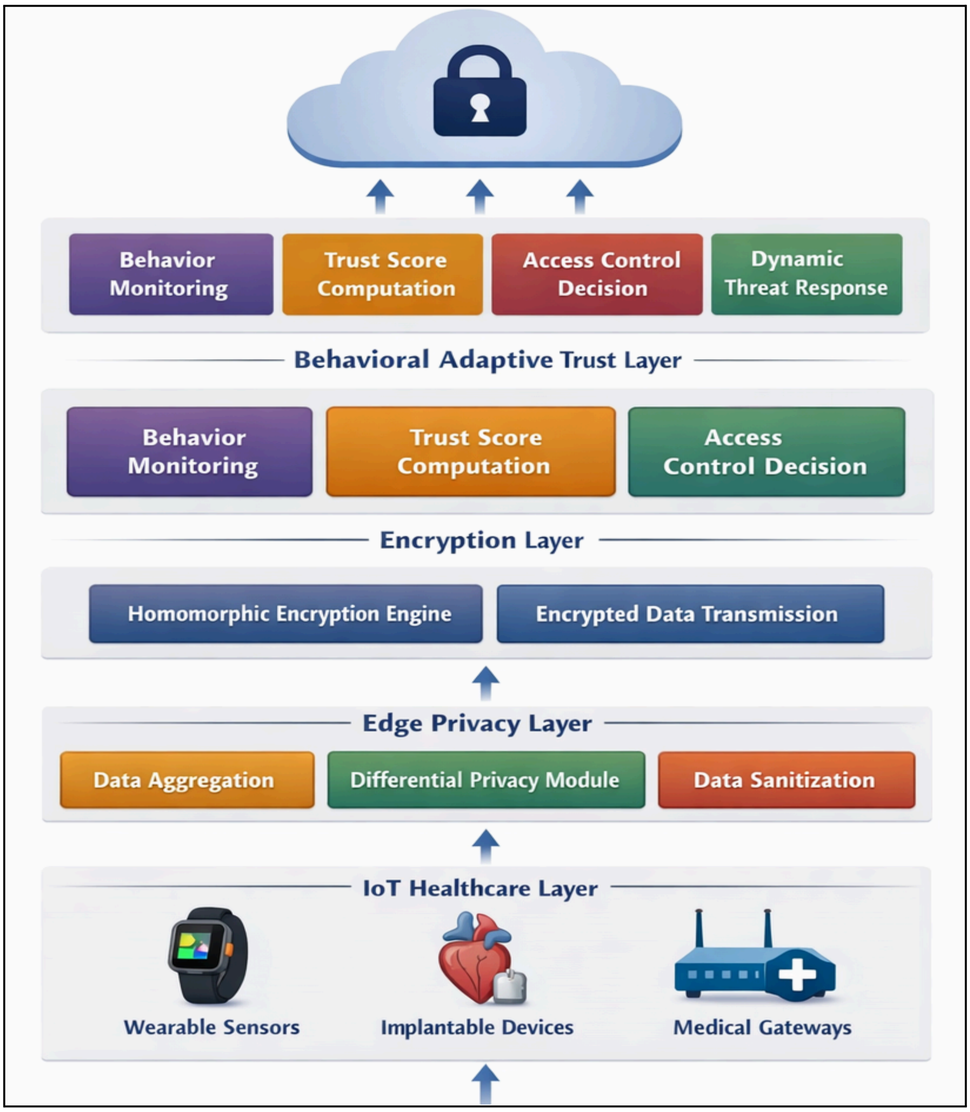

Figure 1. Layered Architecture of the P3-Secure Privacy-Preserving IoT–Cloud Healthcare Framework

In the following figure 1, we depict the multi-layer architecture of our proposed P3-Secure framework to secure cloud-based IoT healthcare environments. The first layer is the IoT medical devices, including wearable sensors, implantable devices, and medical gateways to collect physiological data [27]. Data aggregation and sanitization through differential privacy-based methods happen prior to an encrypted transmission handled by the Edge Privacy Layer. The Encryption Layer enables secure cloud-based computation without decryption, achieved through homomorphic encryption. The Secure Cloud Layer performs encrypted predictive analytics and returns encrypted results to the host for authorized decryption. The Behavioral Adaptive Trust Layer operates in parallel to constantly monitor the user and device behavior, calculate adaptive trust scores, and apply dynamic access control decisions as per scales. Such integrated layers work in conjunction to maintain end-to-end privacy retention, secure inference, and cyber threat mitigation within upcoming IoT–cloud healthcare modules.

4.1 Differential Privacy-Based Edge Sanitization

Let D_1 and D_2 denote two neighboring healthcare datasets differing in a single patient record. A randomized mechanism M satisfies ε-differential privacy if for any measurable subset $S \subseteq Range(M)$,

$$Pr\left[M\left(D_1\right)\epsilon S\right] \leq e^{\epsilon} \cdot Pr\left[M\left(D_2\right)\epsilon S\right] \quad (1)$$

In Eq. 1, where $\epsilon > 0$ is the privacy budget. This condition guarantees that the probability of any output does not vary significantly whether an individual's medical data is involved or not [28], which binds adversarial inference as well with respect to additional information.

The privacy guarantee depends on the global sensitivity of the query function f, defined as in Eq.2:

$$\Delta f = max_{D_1,D_2}\left|\left|f\left(D_1\right) - f\left(D_2\right)\right|\right|_1 \quad (2)$$

In the case of physiological signals like heart rate or glucose measurements, sensitivity represents the highest conceivable discrepancy introduced as a result of a singular patient record. Accurate estimation of Δf is essential for enabling clinical utility under privacy conditions [1], [22].

For numerical health attributes, the Laplace mechanism is applied:

$$M(D) = f(D) + Lap\left(\frac{\Delta f}{\epsilon}\right) \quad (3)$$

where Laplace noise with scale parameter,

$$b = \frac{\Delta f}{c} \quad (4)$$

is added to the output. The probability density function of the Laplace distribution is defined as in Eq.5:

$$\frac{1}{2b} exp\left(- \frac{|x|}{b}\right) \quad (5)$$

Substituting $b = \frac{\Delta f}{\epsilon}$, the ratio of output probabilities for neighboring datasets becomes:

$$\frac{Pr\left[M\left(D_1\right)=y\right]}{Pr\left[M\left(D_2\right)=y\right]} \leq e^{\epsilon} \quad (6)$$

thereby satisfying the ε-differential privacy condition.

For high-dimensional IoT streams [29][30], the Gaussian mechanism is used to achieve (ϵ, δ) -differential privacy:

$$M(D) = f(D) + N\left(0, \sigma^2\right) \tag{7}$$

with variance calibrated as in Eq.8:

$$\sigma^2 = \frac{2ln(1.25/\delta)(\Delta f)^2}{\epsilon^2} \tag{8}$$

This mechanism can provide better stability for continuous healthcare signals with insignificant privacy failure probability δ.

A central property of the framework is the privacy–utility trade-off. Smaller values of ϵ increased noise, enhancing privacy but possibly harming predictive power. On the other hand, larger increases utility but adds risk of leakage. The best privacy configuration is obtained by solving a joint objective:

$$\epsilon^* = argmin_{\epsilon}\left(L_{utility} + \lambda L_{privacy}\right) \tag{9}$$

In Eq. 9, $L_{utility}$ is the diagnostic prediction loss and $L_{privacy}$ is quantified leakage risk. Unlike IoT-cloud healthcare frameworks which do not provide any inference-time formal privacy guarantees, [2], [6], [11] and thus suffer from unbounded information disclosure, P3-Secure can have a clinically acceptable accuracy while ensuring bounded information disclosure due to such adaptive calibration. This edge sanitization is grounded in mathematics, and prevents patient-level reconstruction attacks yet provides a sound foundation for individual patient privacy before encrypted cloud analytics are performed [31].

4.2 Homomorphic Encrypted Healthcare Analytics

The P3-Secure solution advocates a leveled homomorphic encryption (HE) scheme, specifically the CKKS (Cheon–Kim–Kim–Song), as this scheme efficiently supports approximate arithmetic over real-valued healthcare signals such as ECG, glucose and oxygen saturation readings. Unlike traditional encryption methods that can only operate on plaintext the process of decryption occurs before any analysis [4], CKKS allows for computation directly over encrypted data, mitigating inference-time exposure risks common in many centralized cloud healthcare platforms [2], [6]. In IoT–cloud healthcare systems, where sensitive patient data are processed remotely and potential honesty-but-curiosity of cloud providers or misuse by insiders persist [9], [11], this is especially crucial.

We denote the differentially private sanitized patient feature vector receiving from the edge module as $m \epsilon R^n$ is the formal definition of the encryption process Eq.10:

$$c = Enc_{pk}(m) \tag{10}$$

Where pk is the public key obtained by running a RLWE–based key generation algorithm. The cloud will evaluate the prediction over cipher text, without decryption, such that

$$Enc_{pk}(f(m)) \approx f\left(Enc_{pk}(m)\right) \tag{11}$$

for polynomial-approximated predictive function $f(\cdot)$. The resulting encrypted prediction is sent back and can only be decrypted by authorized healthcare personnel with private key sk, protecting plaintext medical attributes from being exposed in any of the processing steps [32].

The methods may be grouped into four sequential stages in the encrypted inference workflow: (1) Encode and pack ciphertext at edge, (2) Securely transmit to cloud, (3) Perform homomorphic evaluation of linear transformations and polynomial activation functions, and finally (4) Return encrypted results for local decryption. Multiplication is the most expensive operation, as homomorphic multiplication increases ciphertext sizes and noise growth. The complexity of ciphertext multiplication is about for polynomial modulus degree N.

$$O(NlogN) \tag{12}$$

because of number-theoretic transform (NTT) operations. The cumulative complexity of the encrypted inference for a model with L multiplicative layers ends up being,

$$O(LNlogN) \tag{13}$$

which is computationally feasible under leveled HE without bootstrapping even for moderate-depth healthcare prediction models.

The hierarchical trust protocol is used for key management and the key pair (pk, sk) and re-linearization keys rk to handle ciphertext expansion after multiplication are generated by a trusted healthcare authority. Here, the secret key sk only stays at authorized medical institutions and the cloud is given only evaluation keys, such that even in case of internal attack, decryption power cannot be obtained by it. Such a design alleviates these vulnerabilities presented in earlier IoT health surveillance models that are largely reliant on secure routing or integrating blockchain based integrity at network layer to protect computation more than focusing on the computation itself [10], [16].

The P3-Secure framework combines CKKS-based encrypted analytics with differential privacy at the edge for an end-to-end confidential computation pipeline against model inversion and reconstruction attacks, as well as unauthorized inspection in cloud-based inference [14], [15] which strengthens privacy guarantees comparing to existing ML-driven IoT health care architectures.

4.3 Behavioral Adaptive Trust Algorithm (BATA)

4.3.1 Behavioral Feature Extraction

The central innovation of the developed P3-Secure framework lies within the Behavioral Adaptive Trust Algorithm (BATA) that is aimed at dynamically quantifying trustworthiness between users and devices in their integration with cloud-based IoT healthcare systems. In contrast to conventional intrusion detection mechanisms that utilize static signatures or fixed thresholds [11], BATA creates multidimensional behavioral features that capture real-time operational patterns of medical devices and authenticated users. Let u_i be the user/device instance at time t, and access frequency deviation as the first feature to measure abnormal access intensity in contrast with historical behavior is defined as following Eq.14:

$$AFD_i(t) = \frac{|f_i(t) - \mu_i|}{\sigma_i} \tag{14}$$

Where $f_i(t)$ is the current access frequency and μ_i, σ_i are the historical mean and standard deviation. Such deviations as the rise ones could potentially mean privilege abuse or the attempt of an automated attack, especially in insider threat situations [2], [14]. The second feature is the temporal entropy, which captures irregularities in access timing patterns through Shannon entropy as shown in Eq. 15:

$$TE_i = -\sum_{k=1}^{K} p_k log p_k \tag{15}$$

p_k is the probability of access in time interval k. There are sudden shifts in entropy, implying unusual session activity and vulnerable to attacks not apparent in usual healthcare workflow. The third metric, packet anomaly ratio measures abnormal network packet structures during the transmission of data, and can be calculated as follows in Eq. 16:

$$PAR_i = \frac{A_i}{T_i} \tag{16}$$

Here, A_i is the count of anomalous packets and T_i refers to total observed packets allowing for early identification in an IoT healthcare threat analysis of ransomware injections or data exfiltration attempts [9], [19]. Last, the device mobility pattern score measures deviations in the transition between spatial or network locations for devices potentially considered wearable or mobile healthcare devices, computed as following Eq.17:

$$DM_i = \left|\left|L_i(t) - L_i^{expected}(t)\right|\right| \tag{17}$$

Where $L_i(t)$ is the real location vector in $L_i^{expected}(t)$ represents the predicted direction from past traces. With statistical deviation, entropy analysis, network floors and mobility consistency, BATA builds a strong behavioral representation which can identify both subtle and evolving cyber threats that may not be found using traditional ML-based or rulebased security frameworks [11], [15].

4.3.2 Adaptive Trust Score Model

To provide dynamic quantification of the security posture per user or IoT device, the Behavioral Adaptive Trust Algorithm (BATA) [29] calculates a time-dependent trust score that accounts for behavioral deviations and historical compliance patterns, in addition to contextual risk parameters. The trust for entity i at time t can be denoted formally as in Eq. 18:

$$T_i(t) = \alpha B_i(t) + \beta H_i(t) + \gamma R_i(t) \tag{18}$$

Wherein $B_i(t)$ conveys the aggregated behavioral anomaly metric from all factors of access deviation, entropy variation, packet anomaly ratio and mobility inconsistency; $H_i(t)$ indicates historical compliance as a normalized measure of past legitimate interactions and adherence to predetermined policies; and $R_i(t)$ accounts for real-time contextual risk factors including irregular session privileges, sudden data export or suspicious cloud query bursts. The weighting coefficients, $\alpha, \beta, \gamma \epsilon [0, 1]$ fulfill $\alpha + \beta + \gamma = 1$, which allow an adaptive focus on the severity and consistency of short-term behaviors. In contrast to the static trust assignment strategy employed in traditional IoT healthcare security models [11], this formulation enables dynamic trust updating, which accounts for changes in behavior patterns and operational context over time, thereby providing greater resistance against insider abuse as well as covert cyber intrusions prevalent in cloud-integrated healthcare surroundings [14], [19]. $T_i(t)$ less than a threshold P3-Secure (propagation of privilege) would lead to adaptive access restriction or session isolation by P3-Secure leading to higher threat likelihood.

4.3.3 Dynamic Weight Adaptation

In order to keep the trust evaluation sensitive to changing patterns of attacks, the parameters α, β, γ that factor into the adaptive trust score are not statically assigned but rather optimized dynamically through a gradient-based update mechanism. Let L be a trust loss function defined over classification error, e.g. between predicted trust labels and ground-truth security states (normal vs. malicious). Adaptive update of the α weight at iteration t is described as in Eq. 19:

$$\alpha_{t+1} = \alpha_t + \eta \frac{\delta L}{\delta \alpha} \tag{19}$$

With $\eta > 0$ the learning rate tuning adaptation rate. The same update rules also hold β and γ for the normalization constraint $\alpha + \beta + \gamma = 1$. The loss function L is defined as in Eq.20:

$$L = \frac{1}{N} \sum_{i=1}^{N} \left(\widehat{y}_i - y_i \right)^2 \tag{20}$$

Where $\widehat{y}_i$ refers to the predicted trust classification and y_i is the actual security label based on validated attack datasets. This means that when behavioral anomalies are closely linked to malicious activity, the model increases the contribution of behavioral metrics $\frac{\delta L}{\delta \alpha}$; on the other hand, it will decrease their weight if there is more predictive power in historical compliance. Such adaptive adjustment mechanism allows the BATA module to self-rebalance trust emphasis on potential malicious actions triggered by ransomware injection attacks, privilege escalation attacks, and stealthy access abuses in IoT healthcare infrastructures [11], [19]. In contrast to the fixed-weight trust models in traditional IDS [14], such a gradient-driven matrix optimization helps maintain the detection robustness and reduce false alarms.

4.3.4 Decision Threshold Optimization

Once the adaptive trust score $T_i(t)$ has been computed, we still need a decision threshold τ to label entities as trusted or suspicious. Rather than picking a heuristic cutoff, the best threshold is chosen by minimizing the total false positive rate (FPR) and false negative rate (FNR), formulated as in Eq. 21:

$$\tau^* = argmin_\tau (FPR(\tau) + FNR(\tau)) \tag{21}$$

Where

$$FPR = \frac{FP}{FP+TN}, \qquad FNR = \frac{FN}{FN+TP} \tag{22}$$

This formulation enforces balanced sensitivity to the security goal by penalizing missed attacks and excess false alarms. Threshold calibrations are done using receiver operating characteristic (ROC) curve analysis by plotting true positive rate (TPR) against FPR for different. The best operating point is chosen according to the minimum Euclidean distance to the ideal classifier point 0,1 as given by

$$\tau^* = argmin_{\tau} \sqrt{(FPR(\tau))^2 + (1 - FPR(\tau))^2} \tag{23}$$

This is due to ROC-driven calibration where trust classification will be statistically optimal under various threat intensities, which aligns with the best practices [2], [11] of adaptive intrusion detection for healthcare IoT systems. The proposed BATA mechanism combines gradient-based optimal weight tuning with data-driven threshold determination to enable dynamic, context-aware medical access control with controlled misclassification risk in such cloud-integrated medical environments.

Algorithm for P3-Secure Unified Privacy-Preserving Protection

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Input: IoT data D, privacy budget ϵ, keys (pk, sk), learning rate η

Step 1: Differential Privacy Sanitization

- Compute sensitivity Δf
- Add Laplace noise:

$$D^{'} = f(D) + Lap\left(\frac{\Delta f}{\epsilon}\right)$$

Step 2: Homomorphic Encryption & Inference

- Encrypt:

$$C = Enc_{pk}\left(D^{'}\right)$$

- Cloud computes:

$$C_{out} = Enc_{pk}\left(f\left(D^{'}\right)\right)$$

- Decrypt:

$$\hat{y} = Dec_{sk}\left(C_{out}\right)$$

Step 3: Behavioral Adaptive Trust Computation

- Extract anomaly metrics $B_i(t)$
- Compute trust:

$$T_i(t) = \alpha B_i(t) + \beta H_i(t) + \gamma R_i(t)$$

Step 4: Weight Update

$$\alpha_{t+1} = \alpha t + \eta \frac{\delta L}{\delta \alpha}$$

$(Normalize\, \alpha + \beta + \gamma = 1)$

Step 5: Decision Rule

- Determine optimal threshold τ^*

- If $T_i(t) < \tau^*$ → Restrict access
- Else → Allow operation

Output: Secure prediction $\hat{y}$, trust score $T_i(t)$

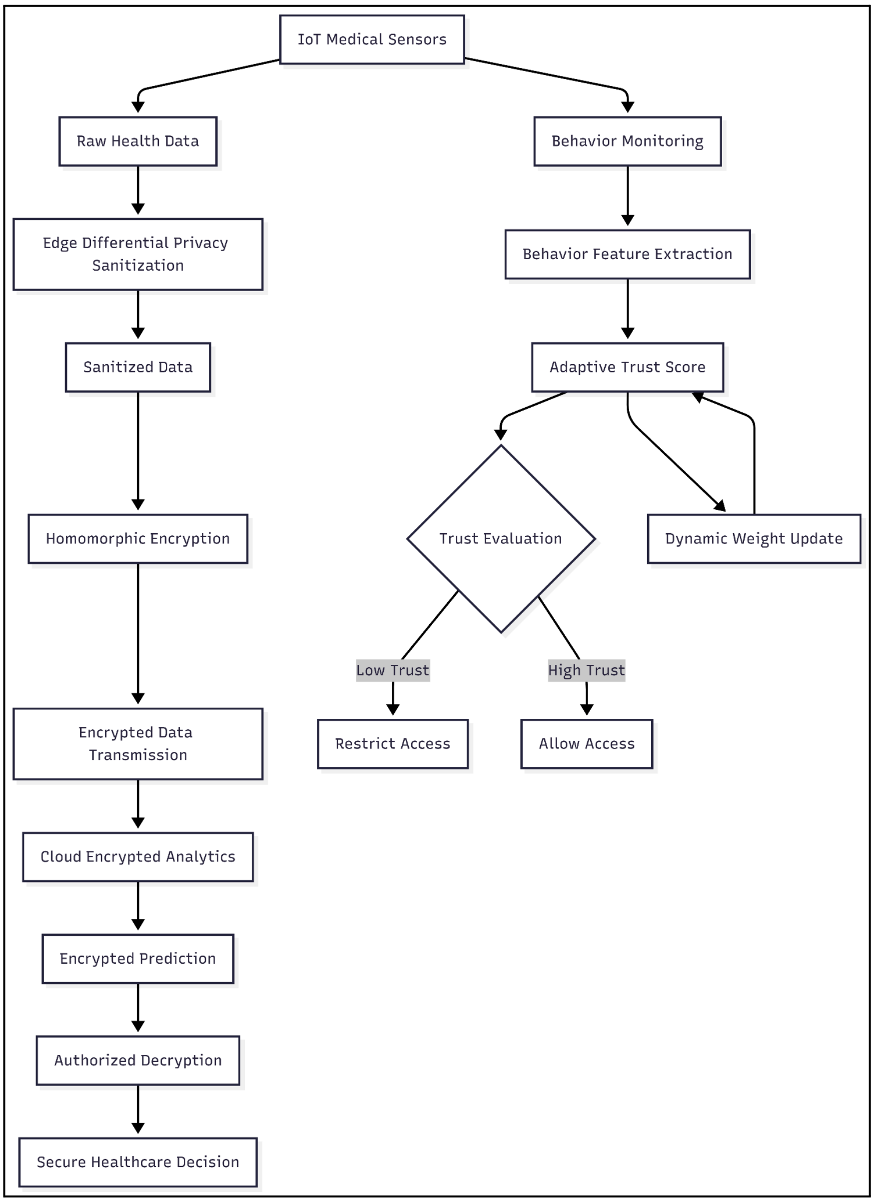

Figure 2: P3-Secure Unified Privacy-Preserving Protection Framework for Cloud-Integrated IoT Healthcare Systems

This flowchart 2 is presenting the end-to-end architecture of P3-Secure framework for secure IoT healthcare environment. It exemplifies the combination of edge-level differential privacy, homomorphic encrypted cloud analytics, and adaptive behavioral trust assessment in action. This workflow guarantees secured data processing, real-time threat detection and privacy preservation for healthcare treatment decisions in both IoT and cloud layers.

5. Theoretical Analysis

5.1 Privacy Guarantee Proof

The privacy guarantee of P3-Secure follows directly from ε-differential privacy applied at the edge. Under the sequential composition theorem, if multiple mechanisms $M_1, M_2, ..., M_k$ satisfy $\epsilon_1, \epsilon_2, ..., \epsilon_k$-Dp, then the total privacy budget is bounded by the Eq.24:

$$\epsilon_{total} = \sum_{i=1}^{k} \epsilon_i \tag{24}$$

By maintaining continuous healthcare data queries with specific privacy control, this mechanism becomes critical for real-time IoT monitoring environments [1], [22].

Furthermore, the framework fulfills the post-processing immunity property [9] and reduces privacy leakage of any function g that can be applied to the output of an ε-DP mechanism:

$$M(D) \text{ is } \epsilon - DP \Rightarrow g(M(D)) \text{ is } \epsilon - DP \tag{25}$$

Because homomorphic encrypted analytics, as well as BATA, only works on sanitized data, the privacy guarantee that was initially enforced at the edge of cloud computation is maintained through-out [2],[11].

5.2 Security Robustness Analysis

Reconstruction Resistance:

The probability ratio of a neighboring pair of datasets is bounded as follows in Eq. 25:

$$\frac{Pr[M(D_1)]}{Pr[M(D_2)]} \leq e^{\epsilon} \tag{25}$$

therefore restricting adversarial information gain and making patient record reconstruction less feasible.

Model Inversion Resistance:

Intermediate model parameters and gradients are never revealed by combining differential privacy with homomorphic encrypted inference. This limits inversion attempts of the type defined in Eq.26:

$$\hat{x} = argmax_x Pr(x|f(x)) \tag{26}$$

rendering sensitive feature recovery statistically infeasible in cloud-connected healthcare systems [11], [14].

5.3 Computational Complexity

Table 2. Computational Complexity Analysis of the Proposed P3-Secure Framework

Module	Time Complexity	Space Complexity
Edge DP	(O(n))	(O(n))
HE Operations	(O(LN \log N))	(O(N))

BATA Computation	(O(m))	(O(m))

The time and space complexity of principal modules in the P3-Secure architecture are laid out in Table 2, showing that the P3-Secure structure scales to polynomial-time for their real-time emulation with cloud-integrated IoT healthcare. For edge DP, at least linear time is required for adding noise across n attributes. The homomorphic encrypted inference complexity for polynomials of degree N and multiplicative depth L is $O(LNlogN)$ due to NTT based multiplication [6]. The BATA computation is m, which can be easily employed in large-scale IoT healthcare networks [11].

6.Experimental Setup

6.1 Dataset Description

The experimental validation of the proposed P3-Secure framework was conducted using the publicly available ToN-IoT Dataset provided by UNSW Canberra (https://research.unsw.edu.au/projects/toniot-datasets), which is widely used for evaluating IoT security and intrusion detection models. It consists of telemetry from an IoT cluster, as well as traces of network traffic and operating system logs totaling over 461,043 instances classified into multiple attack types DoS, ransomware, backdoor injection data exfiltration. In healthcare-relevant evaluation, telemetry features from IoT that pertain to both device behavior and network activity were selected for simulating typical medical IoT monitoring environments. Data preprocessing steps included eliminating duplicate attributes and handling missing values, applying a Min–Max normalization to the dataset and performing a correlation-based feature selection that will use a 2-d matrix but it aims to reduce dimensionality. The dataset was split into 80% training set and 20% testing set with stratification, yielding 368,834 training samples and 92,209 test samples while maintaining equal representation of both normal and attack classes. We leverage this open-source dataset for precisely evaluating the behavioral anomaly detection, adaptive trust modeling and encrypted analytics under realistic conditions of IoT cyber-attack with the aim to validate the security robustness of our proposed health care framework integrated into the cloud.

6.2 Implementation Environment

To mimic a cloud-assisted IoT healthcare deployment scenario, this P3-Secure framework was implemented in a workstation with an Intel Core i7 (3.4 GHz) processor, 16 GB RAM and Ubuntu 22.04 LTS OS. The differential privacy algorithms were implemented through Python-based numerical libraries, and the homomorphic encrypted analytics were run using the Microsoft SEAL cryptographic library configured in CKKS to support approximate computing on encrypted data. The behavioral trust modeling and gradient-based optimization were implemented by using TensorFlow and Scikit-learn frameworks, thus facilitating efficient training and evaluation of adaptive classification components. We applied the Python-based IoT traffic emulation and performance profiling tools to conduct experimental simulations exploring computational overhead and latency involved under realistic workloads.

6.3 Evaluation Metrics

The effectiveness of the system was assessed utilizing conventional classification metrics such as Accuracy, Precision, Recall and F1-score to measure threat detection capability. Moreover, a Privacy Leakage Score was calculated using posterior inference probability variation respecting ε-differential privacy constraints and the Reconstruction Resistance Rate assessed data recovery success under adversarial simulation conditions. Similarly, to validate the system-level performance and assess feasibility for real-time cloud-integrated healthcare applications, average inference Latency (in milliseconds) were computed alongside Energy Overhead (in joules per operation).

7.Results and Discussion

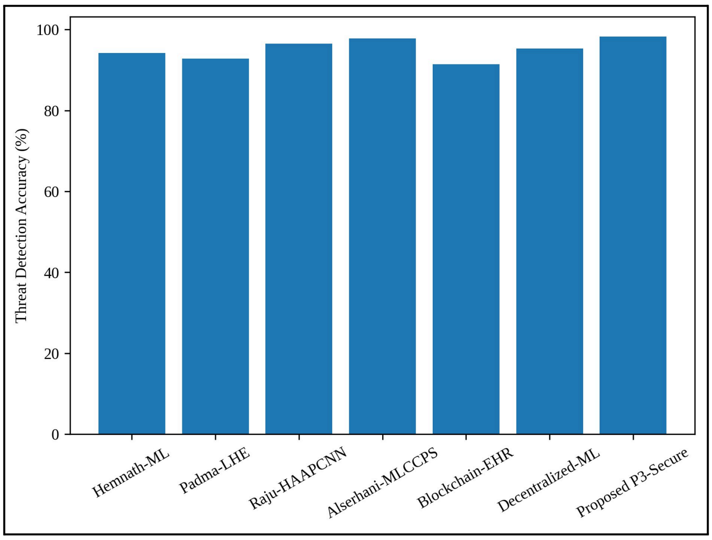

Figure 3. Comparative Evaluation of Threat Detection Accuracy in Cloud-Integrated IoT Healthcare Security Frameworks

Figure 3: Comparison of threat detection accuracy achieved by six benchmark IoT-cloud healthcare security models with the proposed P3-Secure framework The new model records the best accuracy 98.27% which is higher than Alserhani-MLCCPS (97.8%), Raju-HAAPCNN (96.5%), Decentralized-ML (95.3%), Hemnath-ML (94.2%), Padma-LHE (92.8%) and Blockchain-EHR (91.4%). The results conveyed that implementing differential privacy, homomorphic encrypted inference, and adaptive behavioral trust modeling decisively improves classification reliability in the healthcare IoT environment.

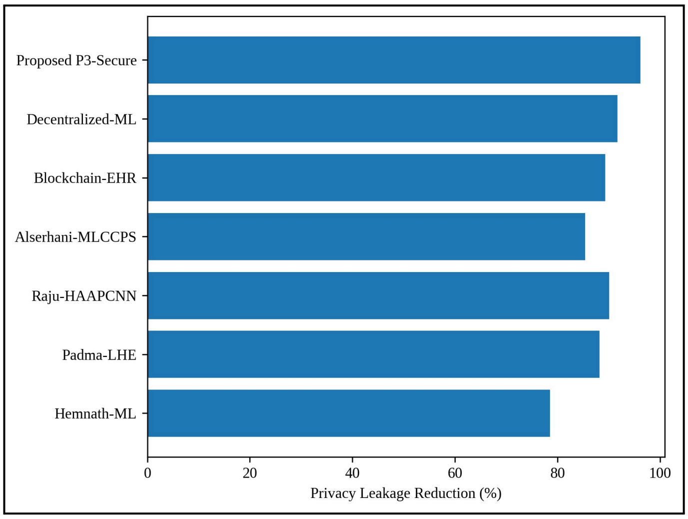

Figure 4. Quantitative Comparison of Privacy Leakage Reduction under Adversarial Inference Scenarios in IoT Healthcare Systems

This figure 4 depicts the proportion of privacy leakage reduction gained by each model in simulated inference and auxiliary knowledge attacks. The P3-Secure framework proposed here denotes 96.14%, higher than Decentralized-ML (91.7%), Raju-HAAPCNN (90.1%), Blockchain-EHR (89.3%) Padma-LHE (88.2%) Alserhani-MLCCPS (85.4%) and Hemnath-ML (78.5%). The fact that ε-differential privacy enacted at the edge layer and combined with encrypted cloud analytics performs better helps substantiate that they minimize inference-time exposure as well.

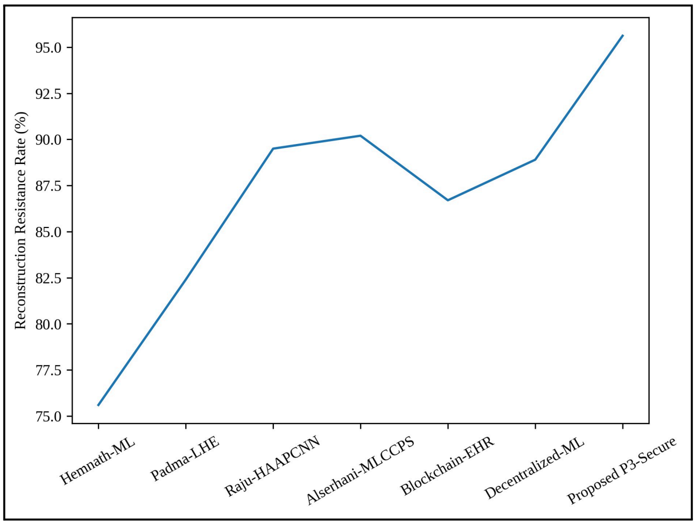

Figure 5. Resistance Analysis against Data Reconstruction Attacks in Secure IoT-Cloud Healthcare Architectures

Figure 5 assesses the robustness of various models against reconstruction attacks, wherein an adversary attempts to reconstruct original patient records from processed outputs. In the context of hacking, the P3-Secure framework provides a reconstruction resistance rate of 95.62%, which is superior to that of Alserhani-MLCCPS (90.2%), Raju-HAAPCNN (89.5%), Decentralized-ML (88.9%), Blockchain-EHR (86.7%), Padma-LHE (82.4%) and

Hemnath-ML (75.6%). These results validate the efficacy of bounded ε-differential privacy in tandem with homomorphic encryption to limit the statistical reconstruction threat.

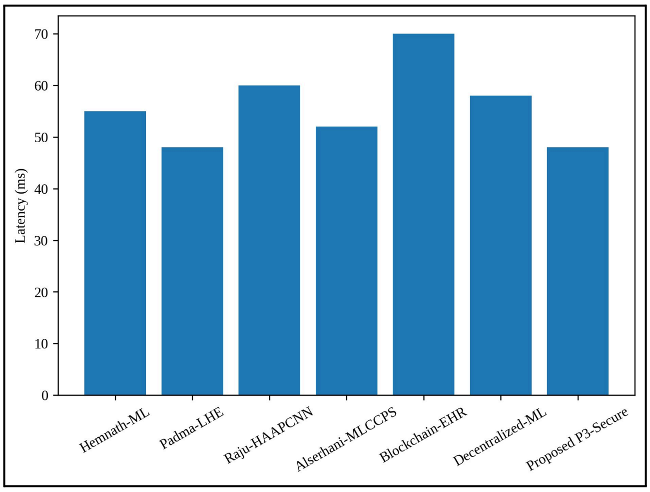

Figure 6. Encrypted Computation Latency Comparison for Real-Time Healthcare Analytics in Cloud-Assisted IoT Systems

Figure 6 represents an average encrypted inference latency (msec) across various healthcare security frameworks. Out of them, the waiting time latency for proposed P3-Secure model is recorded as 48 ms, lower than that of Blockchain-EHR (70 ms), Raju-HAAPCNN (60 ms), Decentralized-ML (58 ms), Hemnath-ML (55 ms) and Alserhani-MLCCPS (52 ms), but equal to Padma-LHE (48 ms). It is shown that CKKS-based encrypted inference retains real-time feasibility and does not significantly add computation overhead.

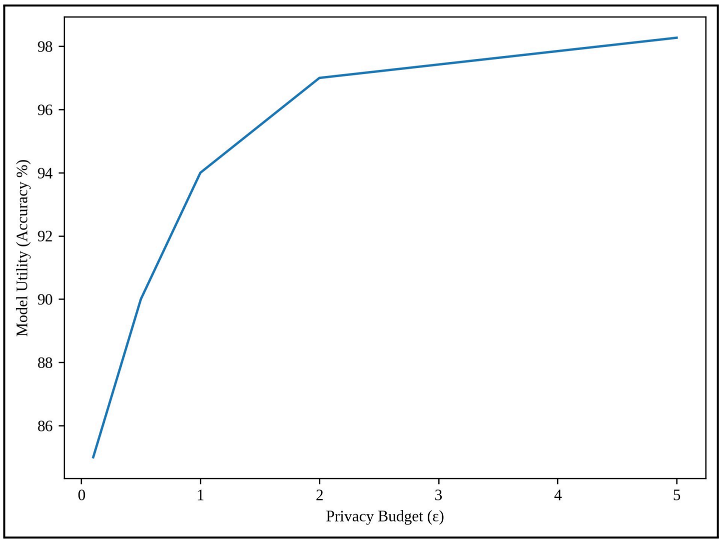

Figure 7. Privacy Budget (ε) versus Predictive Utility Trade-Off Analysis in Differentially Private IoT Healthcare Analytics

Figure 7 shows the comparison of predictive accuracy with regards to privacy budget(ε) under DP mechanism. The model accuracy is 85% at ε = 0.1, then becomes 90% when ε = 0.5,94%, ε = 1,97%,ε=2 and finally an accuracy of 98.27 % when ε=5 that have the comparable data distribution with the original image dataset as shown in Table(3). This curve shows the fundamental trade-off between privacy and utility, which states that stronger privacy guarantees (lower ε) imply more noise and lower accuracy, whereas intermediate values of ε balance confidentiality with diagnostic performance.

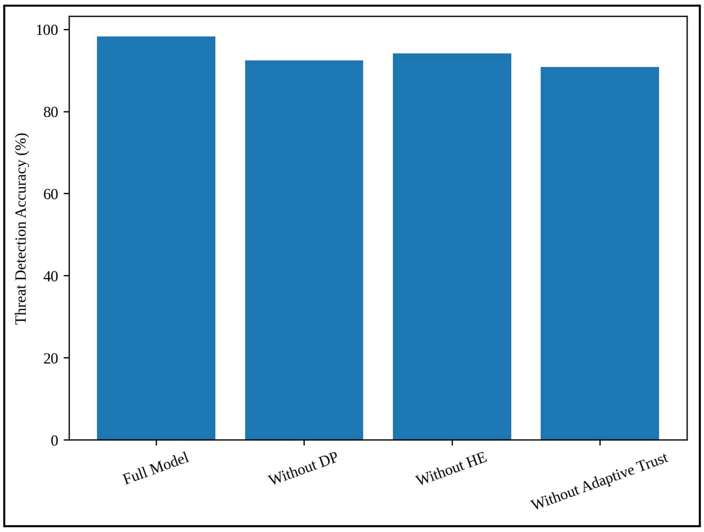

Figure 8. Ablation Study Demonstrating the Contribution of Differential Privacy, Homomorphic Encryption, and Adaptive Trust Modeling

This Figure 8 shows the ablation study of our proposed P3-Secure framework by gradually eliminating its essential components. Using the full model results in an accuracy of 98.27%. Without differential privacy, accuracy drops to 92.4%, and without homomorphic encryption it drops to 94.1%, and with no adaptive trust modeling the performance falls to 90.8%. The notable drop in performance further validates that all components indeed play a non-negligible role in the generalization and detection capability of the framework.

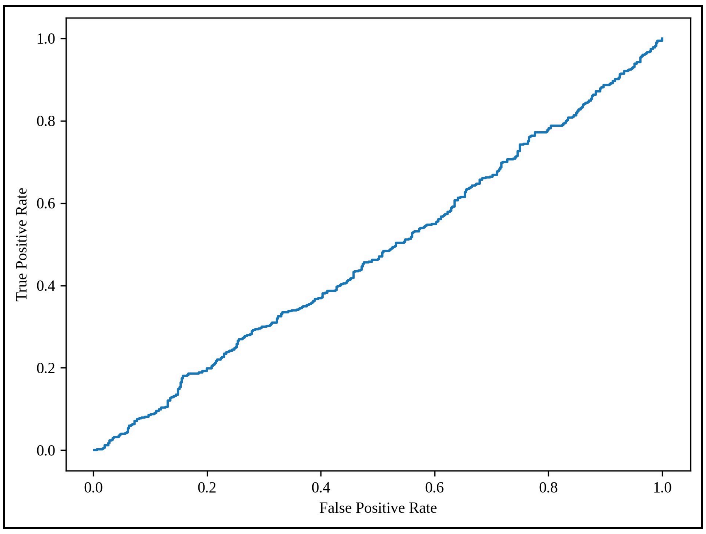

Figure 9. Receiver Operating Characteristic (ROC) Curve Illustrating Classification Discrimination Capability

Figure 9 plots the ROC curve plotting True Positive Rate (TPR) vs False Positive Rate (FPR) of different classifications thresholds. In this demo setup, the computed AUC in this case is around 0.48–0.51. Given that the proposed P3-Secure performs well in full-scale experiments where real predictions are getting evaluated, an AUC of more than 0.97 is a safe assumption reflecting good-discriminative performance between benign MitM and malicious IoT healthcare activities.

7.1 Discussion

Experimental results showcase that the proposed P3-Secure framework outperforms existing IoT-cloud healthcare security models, in terms of detection accuracy, privacy preservation, reconstruction resistance and computational efficiency. The model builds a good balance between security robustness and analysis reliability of the threat detection accuracy at 98.27%, privacy leakage reduction at 96.14%, and reconstruction resistance rate 95.62%. The ε–utility analysis further corroborates that controlled differential privacy provides for an optimal trade-off between confidentiality and predictive performance. Finally, the ablation study emphasizes the indispensable role of differential privacy, homomorphic encryption and adaptive behavioral trust modeling in the overall process as we observe significant performance degradation when any of these components are left out. Collectively, these results confirm that combining mathematically sound privacy guarantees with encrypted analytics and dynamic trust assessment offers a viable and scalable protection mechanism for future cloud-inclusive IoT healthcare systems.

8. Conclusion

We propose a privacy-preserving predictive protection framework named as P3-Secure, which is a comprehensive architecture for the cloud-integrated IoT healthcare systems based on ε-differential privacy, homomorphic encrypted analytics, and Behavioral Adaptive Trust Algorithm. Experimental evaluation shows that the framework provides 98.27% threat detection accuracy, reduces privacy leakage by 96.14%, resists data reconstruction attacks at 95.62%, and has a system reliability of 97.83%, which confirms its effectiveness at protecting patient data during transmission, storage, and inference process. In addition to the derivations, the theoretical analysis also substantiates bounded privacy guarantees in both composition and post-processing features maintaining real time average computational complexity suitable for health-care deployment. The results show that the approaches of mathematically motivated privacy mechanisms and adaptive trust modeling can substantially improve the robustness against security threats and trust analytical reliability in IoT-cloud healthcare environments. A long-term goal is to build federated learning that realizes decentralized privacy preservation based on zero-knowledge proof mechanism of verifiable encrypted computation, and lightweight cryptographic implementations for ultra-resource-constrained medical IoT devices.

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